

Templar Protocol Litepaper

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Abstract

The Templar Protocol is a decentralized modular lending protocol that provides overcollateralized loans. It will be built on the NEAR Protocol, a blockchain known for its scalability, reliability, and Chain Abstraction tools.

Key features of the Templar Protocol include the provision of loan with configurable interest rate, repayment duration, collateral ratio, borrowing limits, supported assets, origination fees, liquidation spread fees, liquidation parameters, insurance fund, and incentives. Importantly, the protocol will also utilize Chain Signatures^[1] to unlock native multichain functionality.

The system's functionality is composed of several operations: borrower operations, lender operations, and liquidation mechanisms. These operations work together to maintain system solvency, align incentives, and ensure the stability of the protocol. To that end, each (collateral asset, borrow asset) liquidity market will be deployed independently to prevent contagion.

Through these mechanisms, the Templar Protocol provides a platform for borrowing against any asset on any chain.

Introduction

Decentralized Finance (DeFi) has created a new ecosystem of financial services that operates without traditional intermediaries. Within this landscape, innovations like smart contracts, and stablecoins have enabled increasingly sophisticated financial applications. The Templar Protocol, introduced in this paper, represents a novel approach bringing decentralized lending markets to assets which don't have native DeFi ecosystems, like Bitcoin. Built on the NEAR Protocol, known for its scalability and user-friendliness, the Templar Protocol aims to provide loans, with a modular design enabling permissionlessly deployed markets. By introducing innovative features such as configurable borrow parameters, a novel yield mechanism, and multichain functionality, the Templar Protocol seeks to become a foundational protocol for multichain, decentralized lending and borrowing.

Key Features of Templar Protocol

- **Overcollateralized borrowing:** Similar to other DeFi lending protocols, Templar allows users to provide assets like Bitcoin as collateral to borrow (usually stablecoins) against without incurring capital gains or missing out on upside potential.

- **Multichain:** The protocol will use Chain Signatures from NEAR's Chain Abstraction stack that will enable users to use assets from other chains as collateral and stablecoin borrows. This is especially useful for ecosystems that do not have native DeFi such as Bitcoin, Ripple, Cardano, Doge, and Litecoin.
- **Modular, Configurable Markets:** Markets are a specific (Collateral asset, Borrow asset) money market. Lenders deposit borrow assets (e.g. stablecoins) and borrowers deposit collateral and receive borrow assets from the market. In addition to configuring privacy and multichain features on market, communities may also specify the following parameters on a per market bases:
 - **Interest Rate:** The Protocol will support variable (based on different utilization curves), constant, and even 0 interest rates.
 - **Loan Duration:** The protocol supports both infinite duration and fixed duration loans.
 - **Liquidation Collateral Ratio:** Static collateral ratio usually determined based on asset volatility allowing borrowers to maximize their borrowing power while also ensuring there is enough buffer in low liquidity conditions so lenders recover their money.
 - **Borrowing Limits:** Minimum and Maximum amount of assets that can be borrowed against per account for a given collateral can be set for each market or asset type to help manage risk and ensure protocol solvency.
 - **Supported Assets:** Supported Collateral and Borrow assets. Each supported asset may have different risk parameters and collateral ratios. New assets can be added permissionlessly as long as liquidity is provided.
 - **Origination Fees:** One-time percent fee charged when a new borrow is created. This is split between the Templar Protocol and the lenders of the market. Templar must receive a minimum of 0.1% of the borrow amount if origination fees are active on a market. This may change based on future governance decisions.
 - **Liquidation Spread Fees:** Fees charged during the liquidation process that represents the difference between the original borrow amount and the value of collateral upon liquidation. This is a [yield mechanism for the protocol](#), complementing interest. It may be distributed among various stakeholders (e.g., lenders, protocol, insurance fund, etc)
 - **Liquidation Parameters:** Parameters relevant to the process for liquidating borrows below the Minimum Collateral Ratio (MCR). Such parameters include transfer of collateral to lenders, full liquidation, or partial liquidation. May include Incentives for liquidators to maintain system health.
 - **Insurance Fund:** Pool of assets set aside to cover potential losses in the system. Can be funded through a portion of liquidation spread fees or other protocol revenues. Provides an additional layer of security for lenders. May be used to compensate users in case of smart contract vulnerabilities or extreme market events.
 - **Supply Incentives:** Rewards or benefits provided to users who supply borrow asset liquidity to the protocol includes a share of the liquidation spread fees and origination fees. May involve governance tokens or community-specific rewards (ie FT tokens, additional, community status/privileges, etc). Designed to attract and retain lenders, ensuring sufficient liquidity in the system.
- **Scalability:** Our system design decisions, including building on top of NEAR protocol, are made with the goal of having 100M+ borrowers and lenders using our protocol. We believe NEAR can scale better than EVM chains^[2] using sharding while providing a superior UX using meta transaction relayers and other unique features of NEAR's account model, and access to other chains using NEAR's Chain Abstraction stack.
- **Security:** We aim to make the design and implementation of the protocol as simple as possible to minimize the attack surface for smart contract exploits. We also designed each market to be independent to prevent cross asset contagion in the event of cascading liquidations and prevent

attacks on a shared liquidity pool by manipulating oracles for low float coins, like what happened to Mango^[3].

- **Growth and Early Adopter Incentives:** The protocol may offer incentives to early adopters and contributors to the system's growth.

Liquidation Spread Mechanism

The Templar Protocol features an approach to generating yield by capturing liquidation spreads. When a borrower's collateral value falls below the required Minimum Collateral Ratio (MCR) threshold, their position becomes eligible for liquidation. During this process, the collateral is sold at market price, which results in some slippage to due liquidity being taken. After the selling is complete, the amount of stablecoins acquired will be somewhere between the original borrow amount, and the price at which the liquidation was originally triggered at, creating a spread. This spread is then captured by the market and distributed among various stakeholders. A portion may go to the liquidators as an incentive for maintaining system health, while another part could be allocated to the insurance fund to enhance protocol stability. The largest percentage of the spread would typically be distributed to lenders as a form of yield. This mechanism allows the protocol to offer lower interest borrows while still providing returns to lenders and maintaining system solvency. The size of the spread serving as yield is influenced by the liquidity for the amount of collateral asset being liquidated, the volatility of the collateral asset, the minimum collateral ratio, and the percentage of the spread distributed to lenders. The percentage of the spread given to lenders vs other parties may be dynamically adjusted based on market conditions, market parameters, or protocol governance parameters, allowing for flexible risk management and yield optimization. The process is visualized in the following flow diagram. Note that all numbers are dollar values for ease of understanding, even if the asset type is NEAR, the collateral asset.

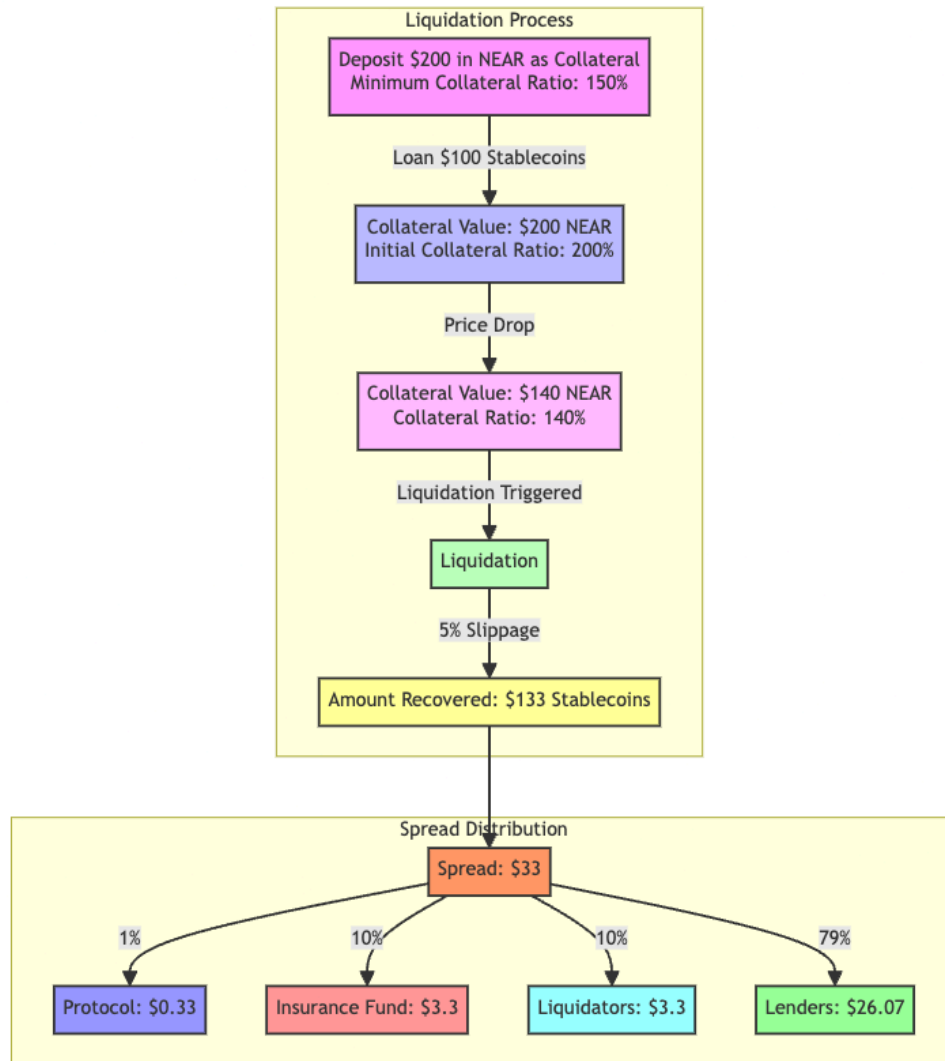


Figure 1

Contract System Functionality

- Borrower Operations:** Borrowers interact with the protocol by connecting an account from their wallet. Once the user finds a suitable market, they deposit a collateral asset (BTC, ETH, NEAR, etc) as collateral and then borrow assets (USDT, USDC, USDS, FRAX, etc) against that collateral. A one time origination fee is charged to the borrower. The Initial Collateral Ratio must be greater than the Minimum Collateral Ratio (MCR) specified for the market. In the future, the MCR may be lowered by staking Templar tokens.
- Supply Operations:** Lenders interact with the protocol by connecting an account from their wallet. Lenders deposit Stablecoins into a market and receive certificates of deposit tokens which map to the specific market and entitle them to a share of the fees and liquidation spread generated by the market or sell on the open market. In the future, Templar tokens would also be issued to lenders which may be staked to earn a share of the protocol fees or sold on the open market.

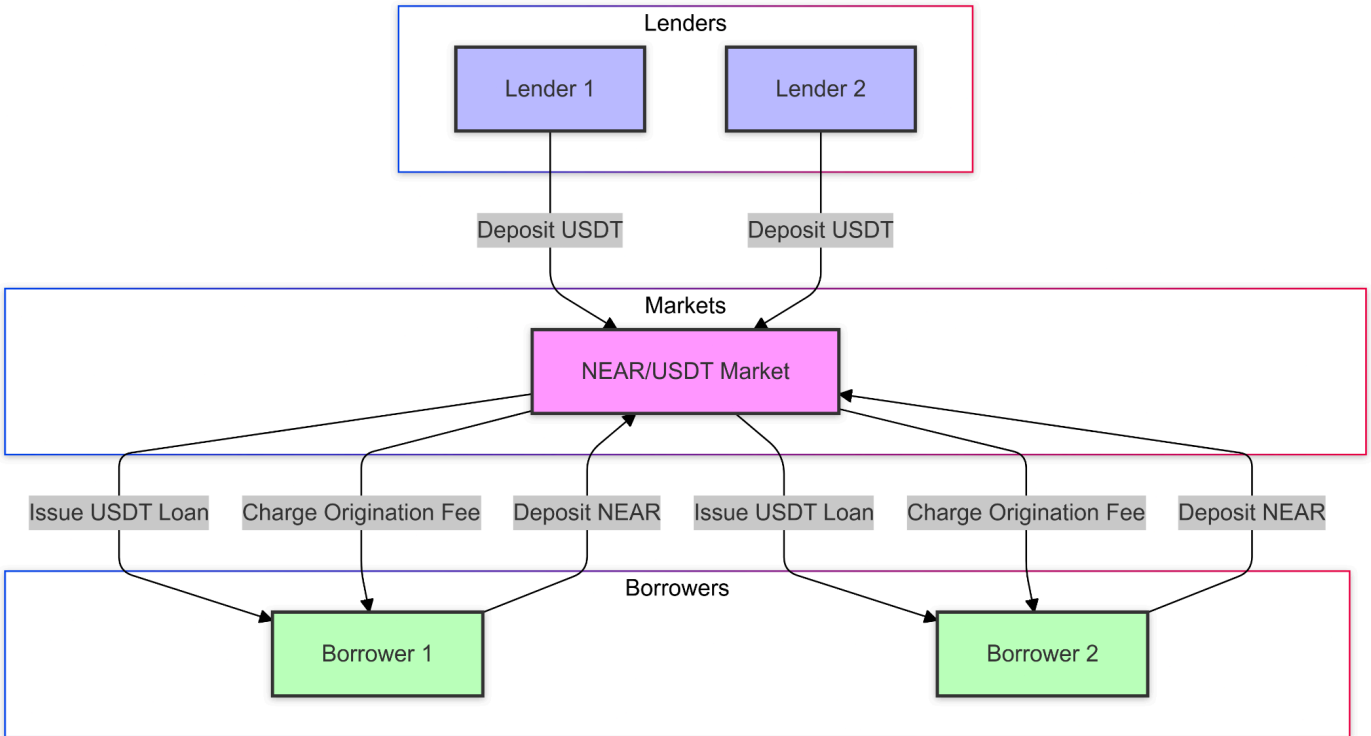


Figure 2

- Borrowing Redemption Mechanism:** At any time, the protocol allows borrowers to repay their borrowed Stablecoins to the market. The borrower may also withdraw their collateral provided the new collateral ratio of the borrow is greater than the minimum collateral ratio + ϵ . If the borrow is fully repaid, then the borrower may withdraw all their collateral.
- Supply Redemption Mechanism:** At any time provided there is ample available liquidity, the protocol allows lenders to withdraw their borrowed stablecoin from the market. Any accrued share of interest, origination fees, token incentive, or liquidation spread yield will also be available for withdrawal. If there are no available stablecoins to withdraw on a specific market, the lender will be added to a queue and the stablecoins will be sent to them automatically when the protocol gets more stablecoins either from more repayments, liquidated collateral, or new lender deposits.

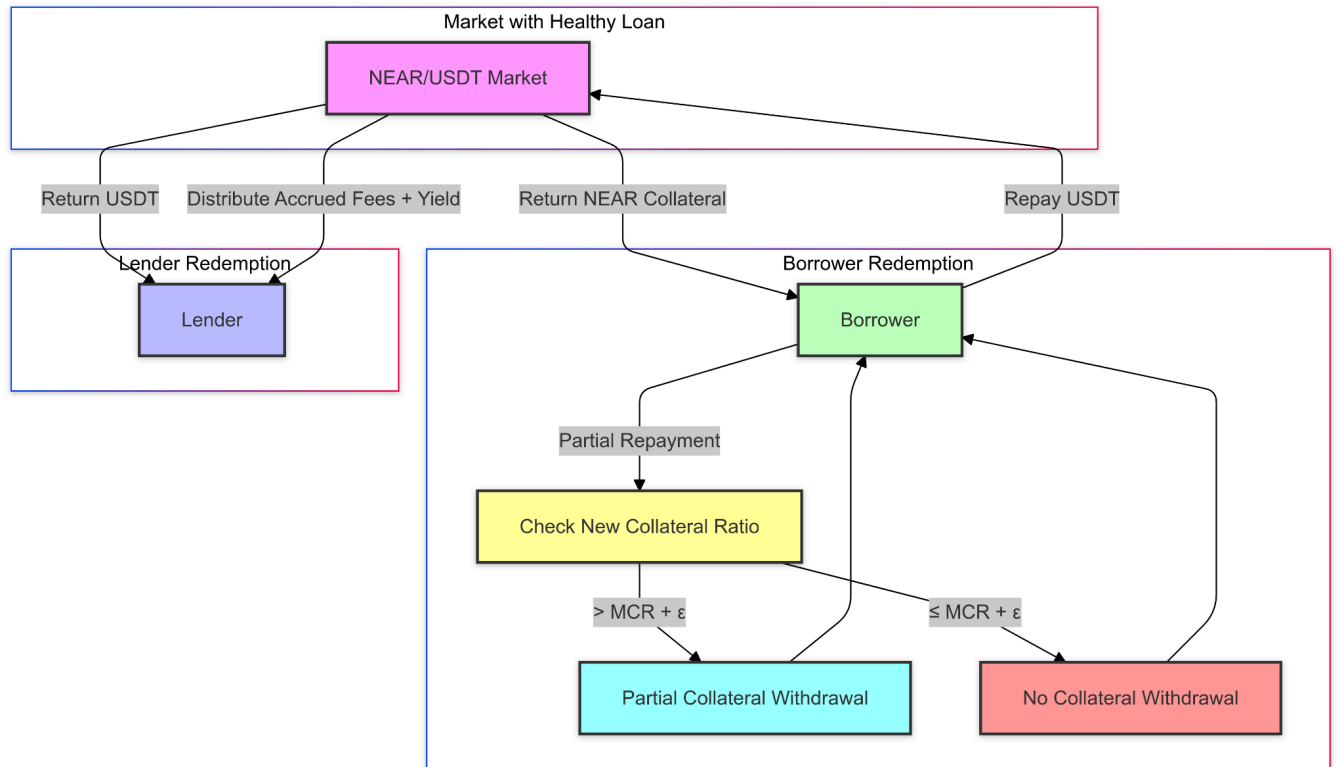


Figure 3

Position Liquidation Mechanism: The liquidation bot iterates through all open borrows in the market, and for all borrows where the current collateral ratio $\leq MCR$, it liquidates the collateral. The liquidation can take several forms depending on the parameterization of the market:

1. Transfer of the collateral asset to lenders who may choose to hold or liquidate
2. Bot fully liquidates the collateral in full into stablecoins
3. Bot partially liquidate the collateral into stablecoins until the collateral ratio for the borrow is above some parameterized threshold

The collateral liquidation may take place on [Near Intents](#), a novel multichain Decentralized Exchange which has access to deep liquidity. This will allow the lenders and the protocol to keep more of the liquidation spread without having to pay liquidators, thus increasing the total yield. Near Intents also removes the need for a liquidation bot as the contract can automatically trigger liquidations. After the liquidation back into stablecoins takes place and slippage is accounted for, any stablecoins in excess of the original borrowed amount will be split between the protocol, liquidators (if not using NEAR intents), insurance fund, and the lenders, with the bulk of the spread going to the lenders. In the future, Templar tokens may be granted upon liquidation to all borrowers, lenders, and liquidators involved.

- **Liquidation Bot:** Off Chain service that tracks open borrows in a market and calls the liquidate function on the market contract on any borrows that are below the MCR. The liquidation bot's account is then awarded a share of the liquidation spread. Lenders, Market Makers, or a subsidiary of the foundation could run the liquidation bots.

- Insurance Fund:** The Insurance Fund takes a share of the liquidation spread. Its purpose is to provide a backstop in the event that the stablecoin value from liquidating collateral is less than the original stablecoin value borrowed. This should only be needed when there are low liquidity conditions, which is most likely to happen during extreme selloffs.

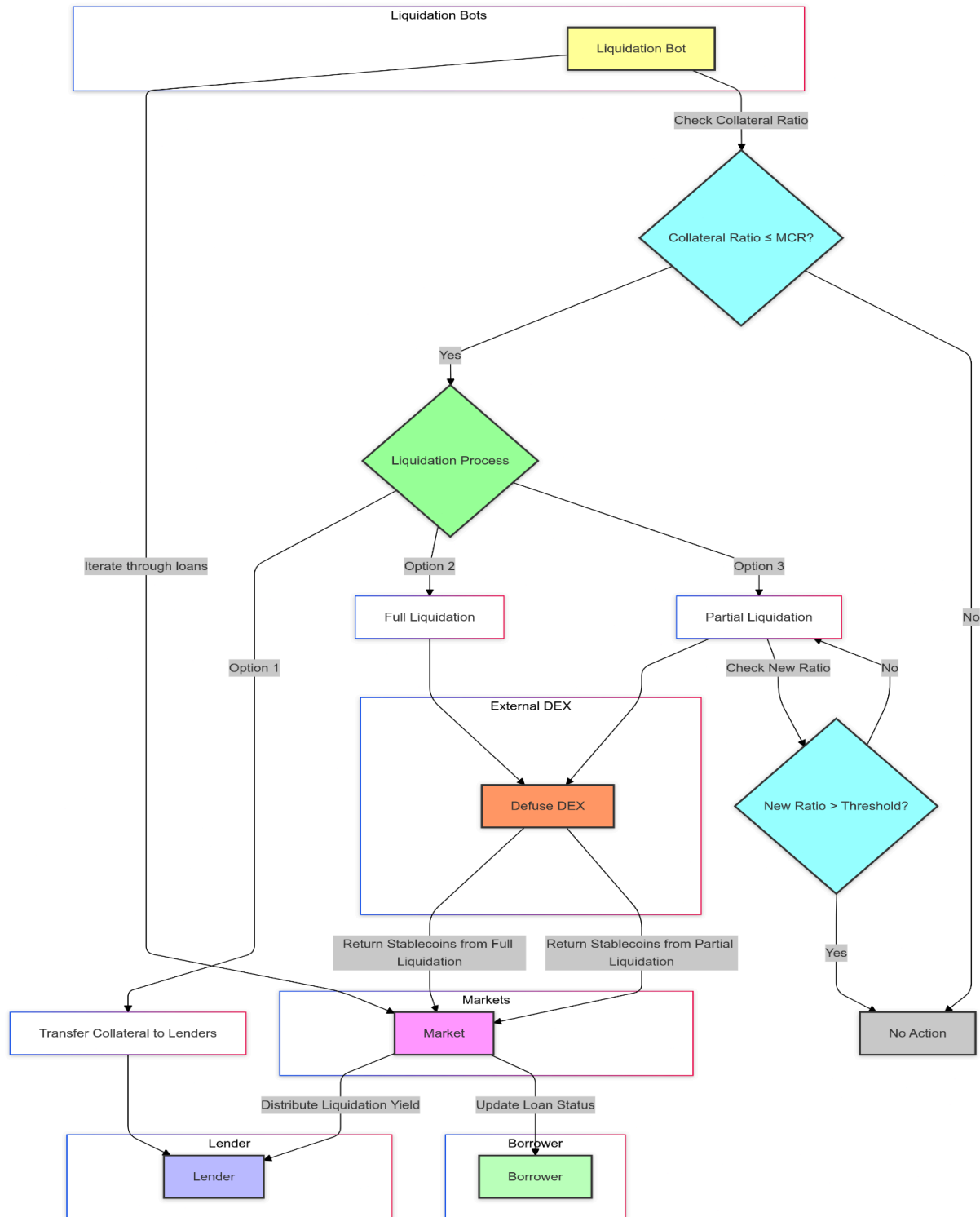


Figure 4

- Recovery Mode:** The protocol recovery mode activates for a market when there are no stablecoins available for borrowers to take out a borrow or stablecoins available for lenders to withdraw. Lenders wishing to withdraw while the protocol is in recovery mode are added to a queue. When it is their turn, the stablecoin will be sent to them automatically when the protocol gets more stablecoins either from more deposits from other lenders, repayments from borrowers, or liquidated collateral from liquidation bots. During Recovery Mode, new borrow creation is disabled. Recovery Mode is deactivated when there are stablecoins available to supply and the recovery mode supply withdrawal queue is empty.

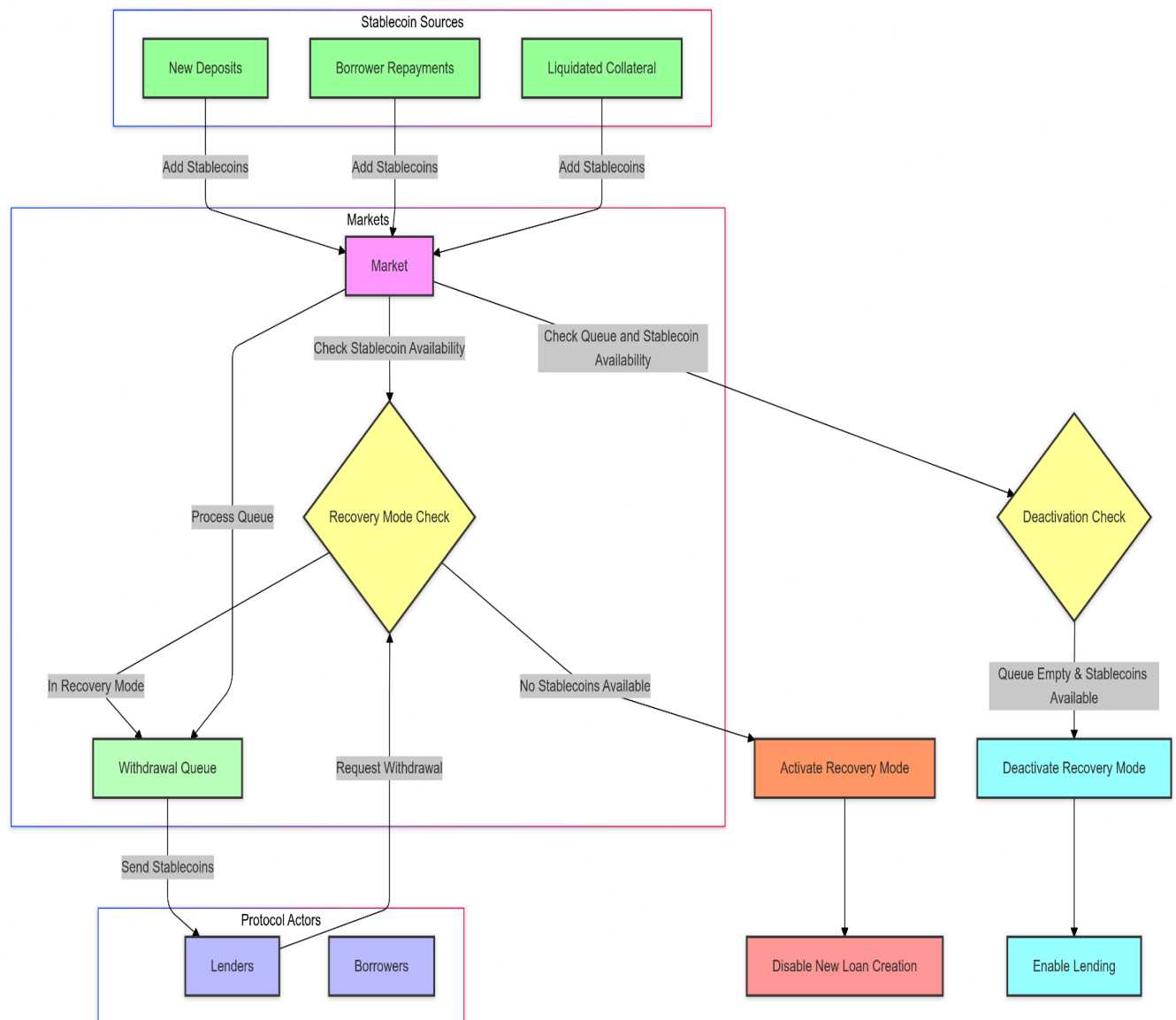


Figure 5

Multichain Functionality

The Templar Protocol leverages NEAR's innovative [chain signatures](#) to extend its capabilities across multiple blockchains, significantly enhancing its interoperability and reach. This integration is crucial for creating a seamless, cross-chain borrowing ecosystem. Here's how the Templar Protocol incorporates chain signatures:

Understanding Chain Signatures

NEAR's chain signatures leverage Multi-Party Computation (MPC) and a distributed network of node operators to create joint signatures from arbitrary payloads, allowing NEAR users to control external blockchain accounts. They also unlock the ability of NEAR smart contracts to sign transactions and the ability to exchange keys that contain balances on foreign chains directly as an alternative to using those keys to sign a transaction on the foreign chain to exchange assets.

Implementation in Templar Protocol

Multichain Collateral

Borrowers can use assets from different blockchains as collateral once they deposit those assets into an account where the keys are controlled via MPC. Chain signatures derive the accounts from the keys and a light client fetches the balance of the assets held by the account.

Multichain Money Markets

The protocol can create borrowing pools that accept borrow and collateral assets from various any blockchain using a secp256k1 or ed25519 curve (all major blockchains).

Chain signatures derive the accounts from the keys and a light client provides the value of assets deposited into the MPC account to the Templar Protocol contract.

Multichain Liquidation Process

Chain Signatures open up more possibilities, and access to broader multichain liquidity, in how collateral liquidations occur. Liquidation bots could:

- a) Create a transaction to liquidate on the native chain
- b) Transfer the keys on NEAR to a Key Exchange for liquidation
- c) Transfer the keys directly to a market subaccount for holding liquidated assets if lenders for a given market want to hold the collateral assets they acquired at a discount during liquidation.

Alternatively, in the future something we can do is replace the binary state machine of "in good standing" and "eligible for liquidation" with 3 states:

1. "in good standing"

2. "needs to be repaid within specific time frame as long as price remains above liquidation collateral ratio"

3. "eligible for liquidation"

In the transition from state 1 to 2, we can alert the user through different means (phone, text, email, app notification) with the details of how much they need to repay or additional collateral they need to provide in order to get the loan back to good standing. Each market would need to be configured properly so that the collateral ratio for state 2 and the liquidation collateral ratio are far enough apart such that the user has ample time to repay.

Benefits of Multichain Functionality

Increased Liquidity

Access to assets and users from multiple blockchains expands the Total Addressable Market of the protocol. It also unlocks liquidity opportunities for large ecosystems that don't have access to native DeFi such as BTC, XRP, DOGE, LTC, etc. In liquidation scenarios, having access to multichain liquidity ensures better execution prices, and thus better spreads and more yield for lenders while allowing potentially lower MCRs for lenders.

Broader Accessibility

Users can participate regardless of their preferred blockchain ecosystem.

Enhanced Risk Distribution

Diversification across multiple chains reduces systemic risks associated with any single blockchain.

Improved Capital Efficiency

Users can utilize their cross-chain assets without needing to bridge them manually.

Conclusion

The Templar Protocol introduces a novel approach to decentralized finance (DeFi) borrowing. Its core innovation lies in a modular architecture for providing chain abstracted stablecoin borrows. This unique model creates a mutually beneficial ecosystem for both borrowers and lenders. The protocol distinguishes itself through several key features:

1. Configurable borrow parameters, allowing for flexibility in collateral ratios, borrowing limits, liquidation thresholds, and more.
2. Yield derived from liquidation spreads, interest, origination fees, and token incentives.
3. Integration with the NEAR Protocol, leveraging its scalability, user-friendliness, and multichain capabilities like Chain Signatures.

These features collectively position the Templar Protocol as a modular and innovative solution in the DeFi space, addressing key challenges of existing borrowing platforms while introducing innovative mechanisms for risk management and yield generation. By combining these elements, the Templar Protocol aims to create a more accessible, efficient, and secure decentralized borrowing ecosystem.

Disclaimer

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